Assignment 2

Experiment Objectives

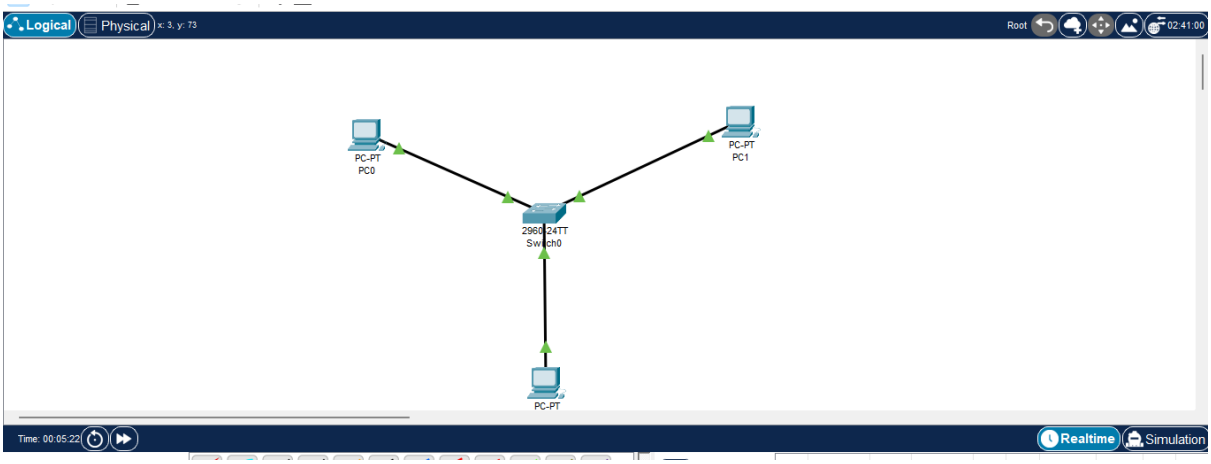
Students will use Packet Tracer Simulation Mode to observe how packets move between devices, how switches learn MAC addresses, and how ICMP echo request/reply travels through a small network.

Aim:- Packet Flow Visualization Using Simulation Mode

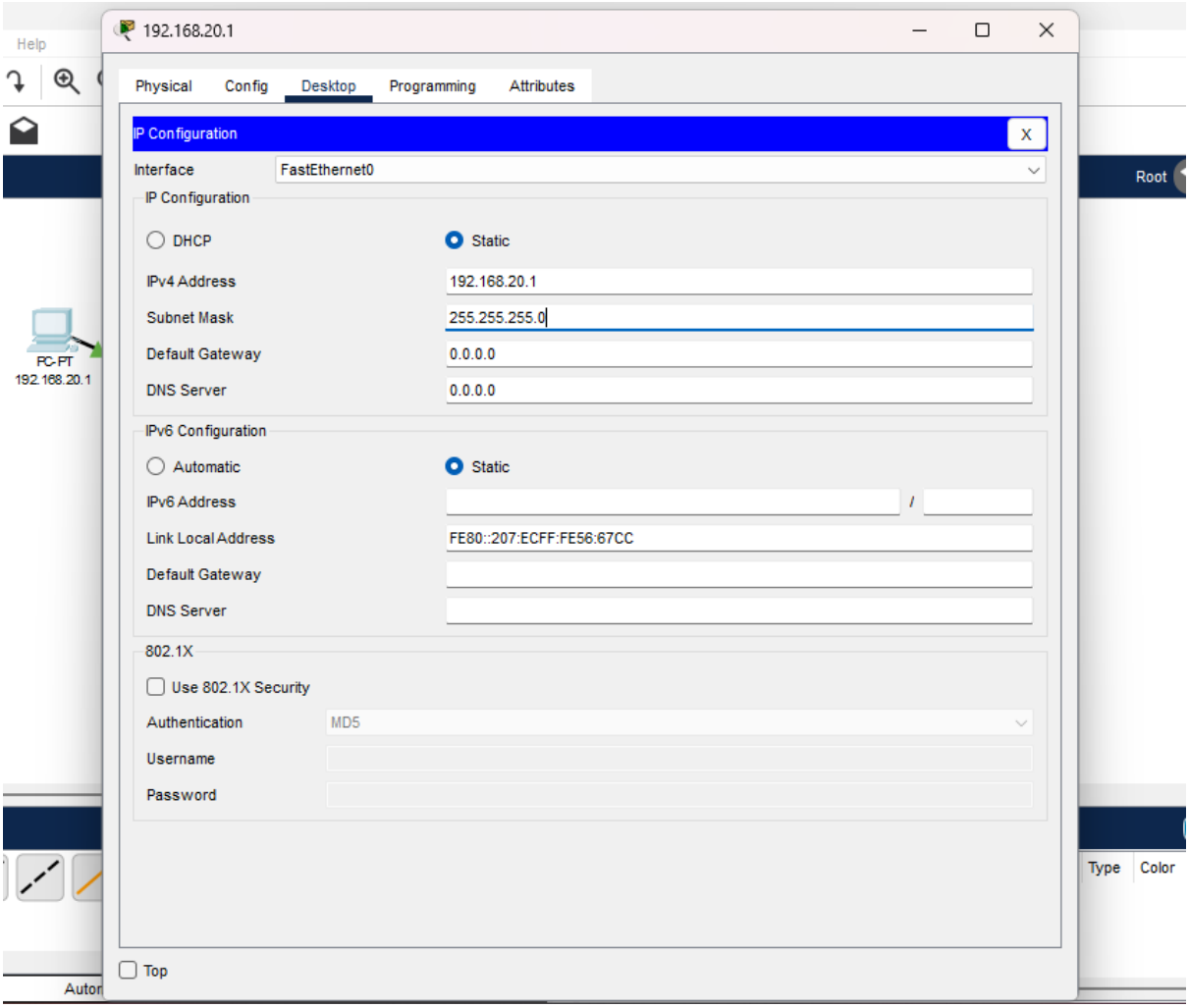
Instructions:-

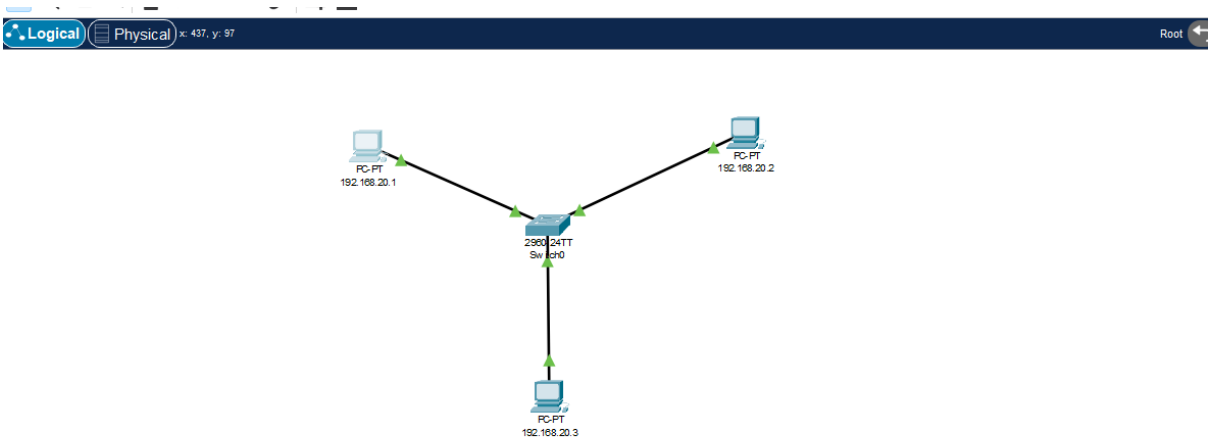
Task 1: Build a Simple LAN

- 1 switch + 3 PCs

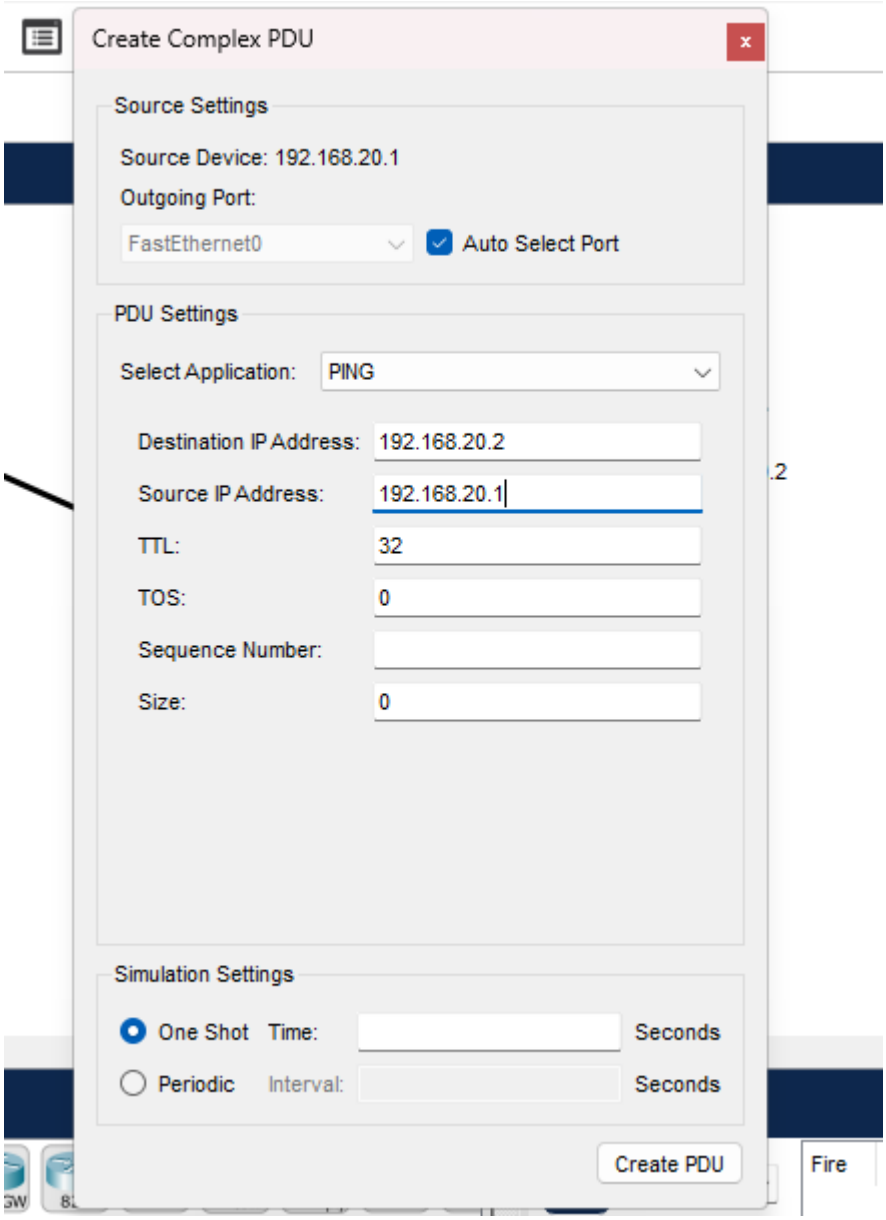


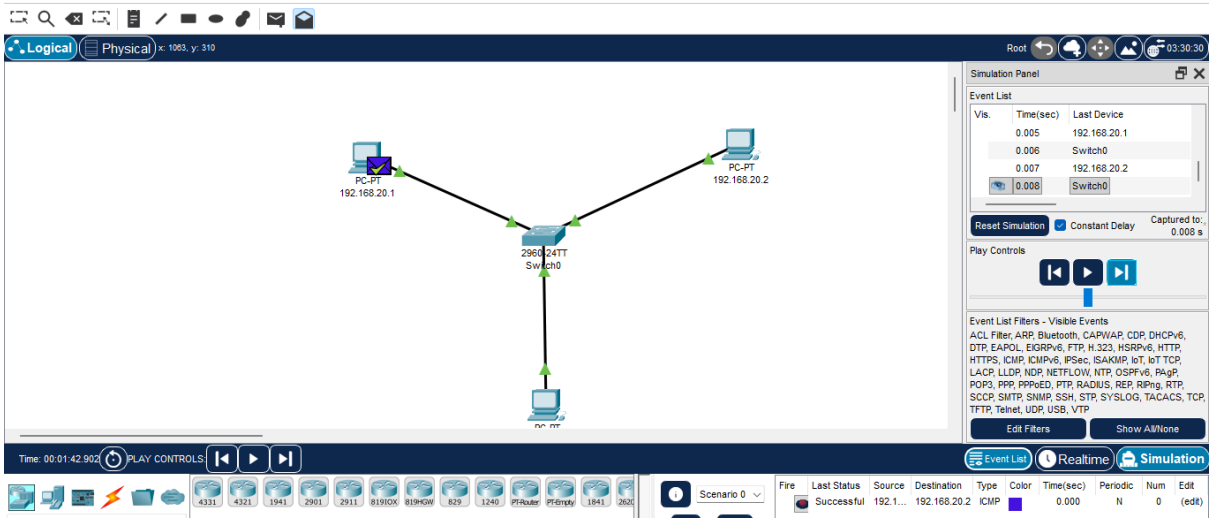
- IP example: 192.168.20.1–3 /24

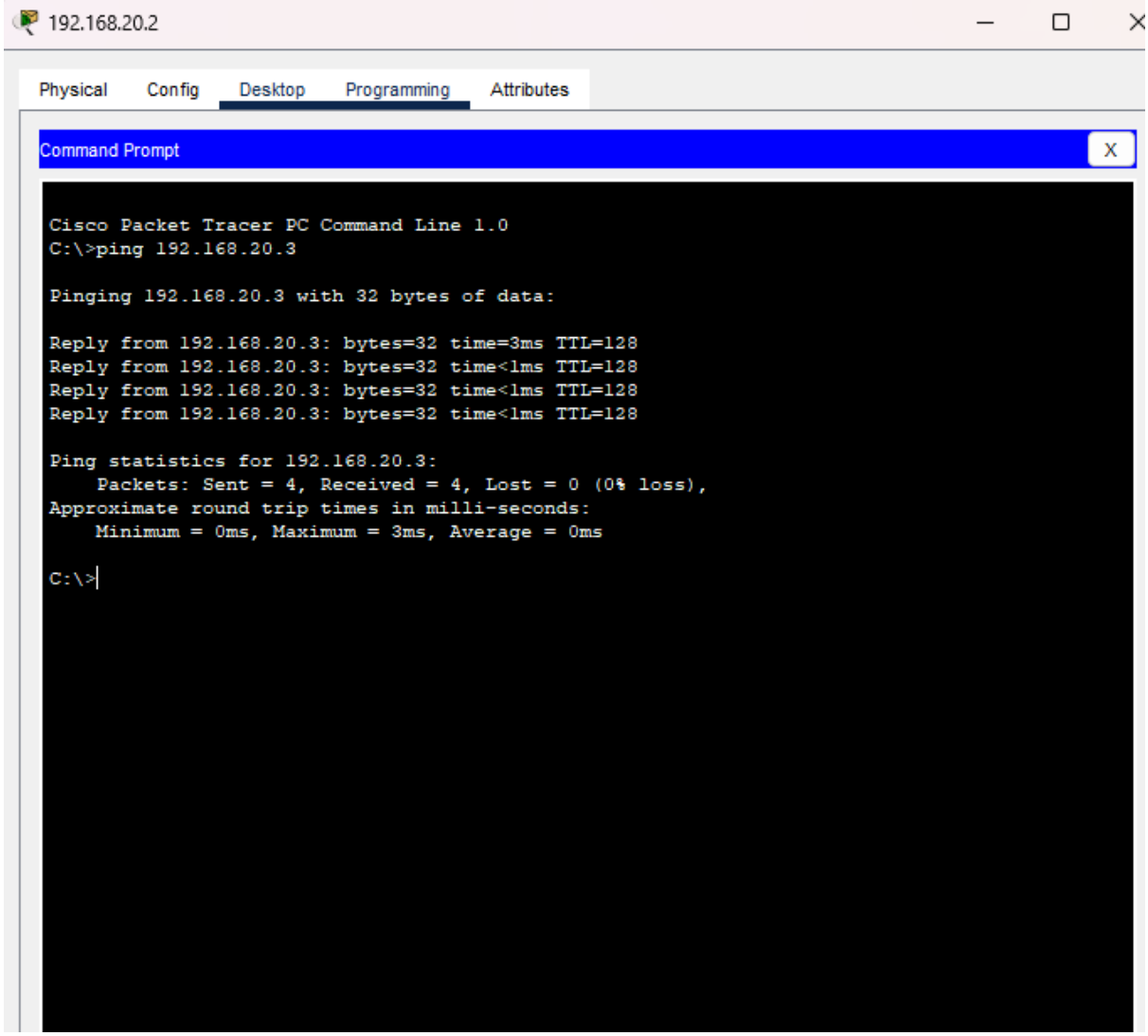




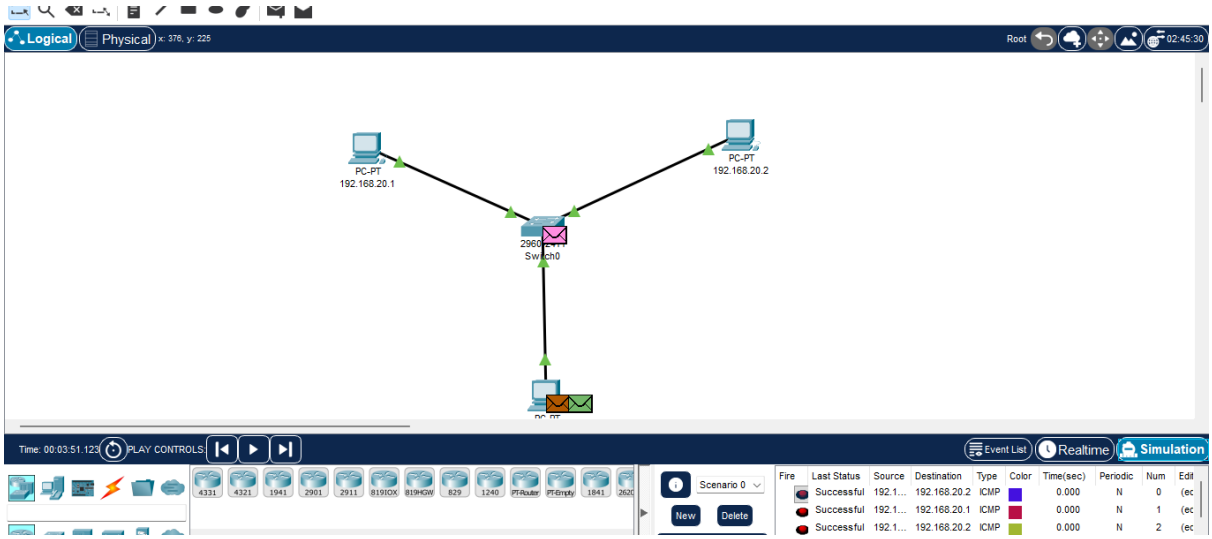
Task 2: First Ping Observation

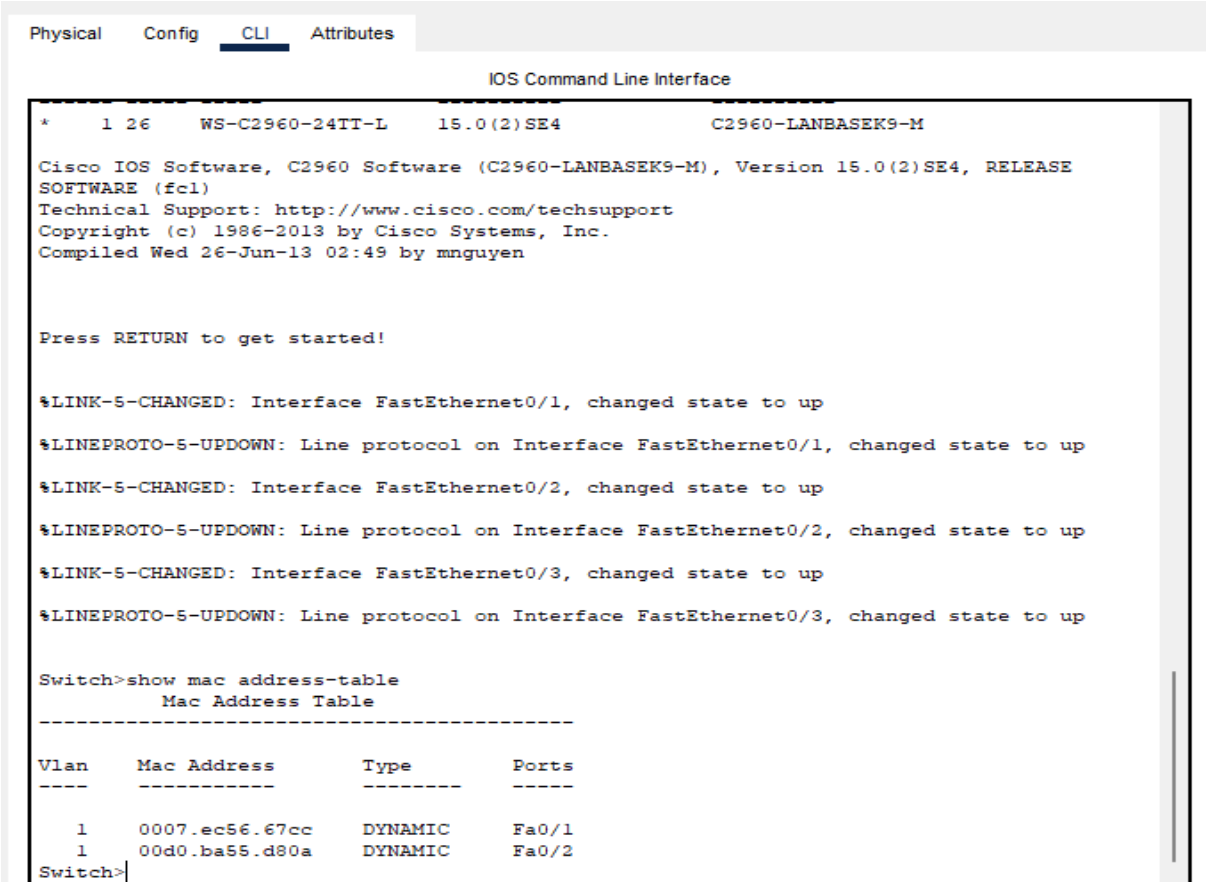
- Enable Simulation Mode
- Ping PC1 → PC2
- Observe ARP request and ARP reply before ICMP
- 


Logical
Physical
PC-PT
192.168.20.1
PC-PT
192.168.20.2
2960-24TT
Switch0
Simulation Panel
Event List
Vis.
Time(sec)
Last Device
0.005
192.168.20.1
0.006
Switch0
0.007
192.168.20.2
0.008
Switch0
Reset Simulation
Constant Delay
Play Controls
Edit Filters
Show All/None
Realtime
Simulation


192.168.20.2
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.20.3
Pinging 192.168.20.3 with 32 bytes of data:
Reply from 192.168.20.3: bytes=32 time=3ms TTL=128
Reply from 192.168.20.3: bytes=32 time<1ms TTL=128
Reply from 192.168.20.3: bytes=32 time<1ms TTL=128
Reply from 192.168.20.3: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.20.3:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 3ms, Average = 0ms
C:\>

Task 3: Second Ping Observation

- Ping PC1 → PC2 again
- Observe fewer packets (ARP cached)
- 

Task 4: MAC Table Check

- View switch MAC table (if available) / observe switch learning through events